

# Lecture 3: Expected Utility

ECON 403/503

Fall 2020

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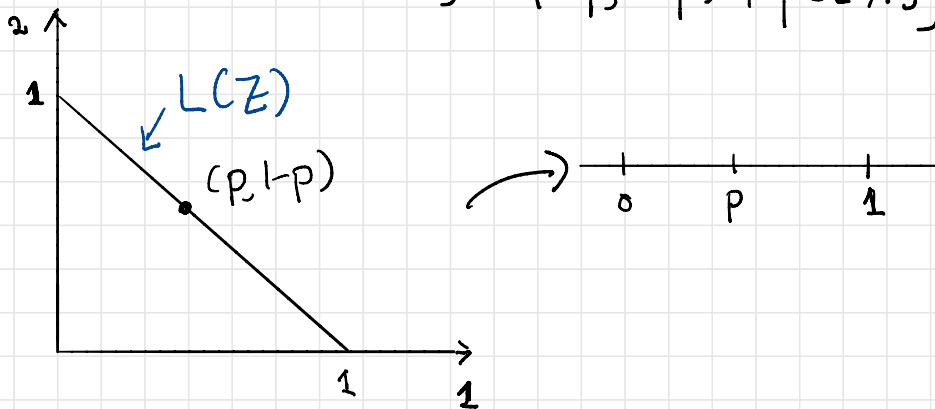


# Understanding the set $L(Z)$

Suppose that  $Z$  is finite.

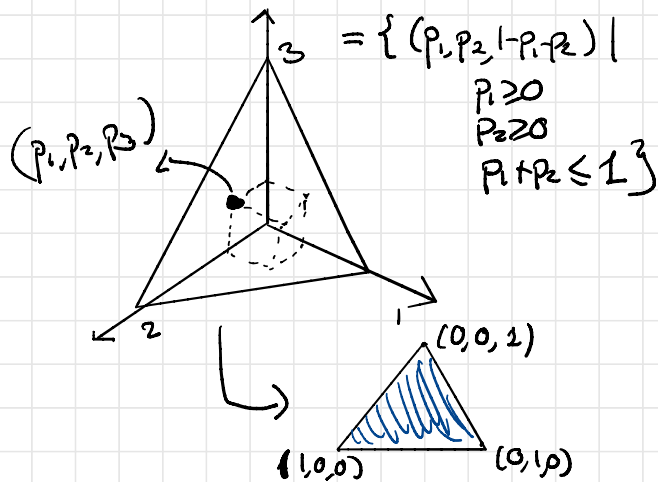
Example 1:  $Z = \{z_1, z_2\}$

$$\begin{aligned} L(Z) &= \{\text{all prob. mass functions over } Z\} \\ &= \{p \in [0, 1]\} = \{(p, 1-p) \mid p \in [0, 1]\} \end{aligned}$$



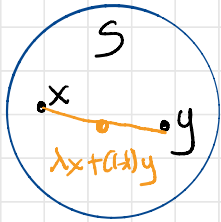
Example 2:  $Z = \{z_1, z_2, z_3\}$

$$\begin{aligned} L(Z) &= \{\text{all prob. mass fn. over } Z\} \\ &= \{(p_1, p_2, p_3) \mid p_1 \in [0, 1], p_2 \in [0, 1], p_3 \in [0, 1], p_1 + p_2 + p_3 = 1\} \end{aligned}$$



## Convexity:

We say that a set  $S$  is convex if for all  $x, y \in S$  and  $\lambda \in [0, 1]$  then  $\lambda x + (1-\lambda)y \in S$ .



convex



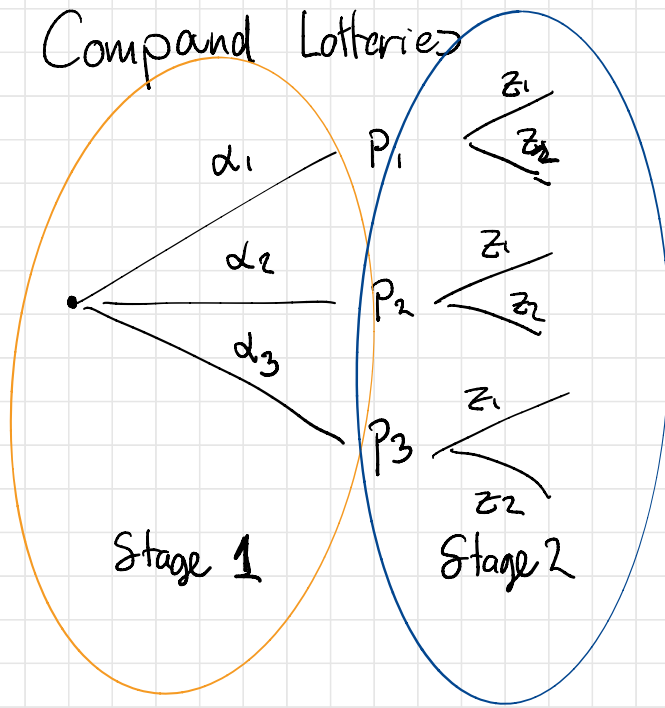
not convex

The set of lotteries  $L(Z)$  is convex.

What happens if  $Z$  is infinite? What is  $L(Z)$ ? To define properly this set you would need measure theoretic concepts.

$L(Z) =$  "set cumulative distribution functions of random variables over  $Z$ "  
 $Z \subseteq \mathbb{R}$

# Compound Lotteries



$$Z = \{\text{Beshbarmak}, \text{Pilaf}, \text{Boursaks}\}$$

$$[\text{Beshbarmak}] > [\text{Pilaf}]$$

If I offer you the following:

① Flip a Coin: If Heads, you eat Beshbarmak  
Tails, you eat Boursaks

② Flip a Coin: If Heads, you eat Pilaf  
Tails, you eat Boursaks

$$① = \frac{1}{2} [\text{Beshbarmak}] + \frac{1}{2} [\text{Boursak}]$$

$$② = \frac{1}{2} [\text{Pilaf}] + \frac{1}{2} [\text{Boursak}]$$

If your preferences satisfy I, then you must

$$\text{satisfy } ① > ②$$

Independence Axiom: also applies to  $\succsim, \sim$

$I^*$ , For every  $(p_k)_{k=1}^K$ ,  $q_{k^*}$  a lottery, and  $(\alpha_k)_{k=1}^K$ ,  $\alpha_k \geq 0$ ,  $\alpha_{k^*} > 0$ ,  $\sum_{k=1}^K \alpha_k = 1$ , then

$$\alpha_1 p_1 \oplus \alpha_2 p_2 \oplus \dots \oplus \alpha_{k^*} p_{k^*} \oplus \dots \oplus \alpha_K p_K$$

$$\succsim$$

$$\Leftrightarrow p_{k^*} \succsim q_{k^*}$$

$$\alpha_1 p_1 \oplus \alpha_2 p_2 \oplus \dots \oplus \alpha_{k^*} q_{k^*} \oplus \dots \oplus \alpha_K p_K$$

Proof: Notice that:

$$\bigoplus_{k=1}^K \alpha_k p_k = \underbrace{\alpha_{k^*} p_{k^*}}_{\text{lottery}} \oplus \underbrace{(1 - \alpha_{k^*}) \left[ \bigoplus_{k \neq k^*} \frac{\alpha_k}{1 - \alpha_{k^*}} p_k \right]}_{\text{lottery}}$$

By Independence

$$\bigoplus_{k=1}^K \alpha_k q_k \stackrel{\text{Independence}}{\Leftrightarrow} p_{k^*} \succsim q_{k^*}$$

$$\downarrow$$

$$\succsim \alpha_{k^*} q_{k^*} \oplus (1 - \alpha_{k^*}) \left[ \bigoplus_{k \neq k^*} \frac{\alpha_k}{1 - \alpha_{k^*}} q_k \right]$$

weight  $\alpha_{k^*}$   
 $1 - \alpha_{k^*}$

$$\bigoplus_{k \neq k^*} \frac{\alpha_k}{1 - \alpha_{k^*}} p_k \sim \bigoplus_{k \neq k^*} \frac{\alpha_k}{1 - \alpha_{k^*}} q_k$$

because  $p_k = q_k$  except  $k^*$   
for all  $k$

$$p_{k^*} \succeq q_{k^*} \Leftrightarrow \alpha_{k^*} p_{k^*} + (1 - \alpha_{k^*}) r \succeq \alpha_{k^*} q_{k^*} + (1 - \alpha_{k^*}) r$$

Independence

where  $r = \bigoplus_{k \neq k^*} \frac{\alpha_k}{1 - \alpha_{k^*}} p_k = \bigoplus_{k \neq k^*} \frac{\alpha_k}{1 - \alpha_{k^*}} q_k$  ( $q_k = p_k$  for all  $k \neq k^*$ )

Monotonicity: comparing  $[x] > [y] \Leftrightarrow \alpha x \oplus (1 - \alpha) y \succeq \beta x \oplus (1 - \beta) y$   
if  $\alpha > \beta$

$$[x] > [y]$$

$$\text{If } \alpha > \beta \Leftrightarrow \alpha x \oplus (1 - \alpha) y \succeq \beta x \oplus (1 - \beta) y$$

Intuition:

The comparison of these two compound lotteries is equivalent to comparing the real numbers  $\alpha$  and  $\beta$

### Example 3.2.

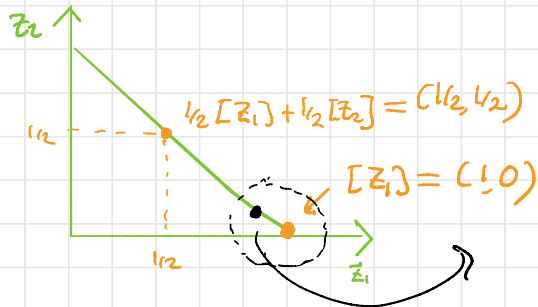
Does not satisfy independence  
nor continuity. To see this  
consider:

Suppose  $[Z_1] > [Z_2]$ , but

$\frac{1}{2}[Z_1] + \frac{1}{2}[Z_2] \sim [Z_2]$ ,  $\therefore$   
doesn't satisfy independence.

We have that  $w([Z_1]) \succ^* w([Z_2])$

but  $[Z_1] > \frac{1}{2}[Z_1] + \frac{1}{2}[Z_2]$



No matter the  
size of  
 $B([Z_1]) = B(1, 0)$

Any lottery  $p \in B([Z_1]) / \{(1, 0)\}$   
will have the property that

$$p \sim \frac{1}{2}[Z_1] + \frac{1}{2}[Z_2] = (1/2, 1/2)$$

Any  $p \in B([Z_1]) / \{(1, 0)\}$

assigns strictly positive probability

to  $Z_1, Z_2$ , therefore,

$$w(p) = Z_2 \prec^* Z_1$$

$$\Rightarrow p \prec [Z_1]$$

### Example 3.3

Satisfies both. This preference can be represented in the following way:

assign value of  $v(z) = 1$

for  $z \in \text{Good}$  and

$v(z) = 0$ , for  $z \in \text{Bad}$ .

Hence  $G(p) = \mathbb{E}[v(z)]$

$$= \sum_{z \in Z} v(z) p(z)$$

$$= \sum_{z \in \text{good}} \overset{1}{v(z)} p(z) + \sum_{z \in \text{bad}} \overset{0}{v(z)} p(z)$$

$$= \sum_{z \in \text{good}} p(z)$$

$$p \succeq q \Leftrightarrow \mathbb{E}[v(z); p] \geq \mathbb{E}[v(z); q]$$

Expected value is:

- Linear  $\Rightarrow$  Independence
- Continuous  $\Rightarrow$  Continuity  
function

### Example 3.4

Doesn't satisfy both.

$[a] \sim [b]$ , but

$0.5[a] + 0.5[b] < [b]$ ,

which violates independence,

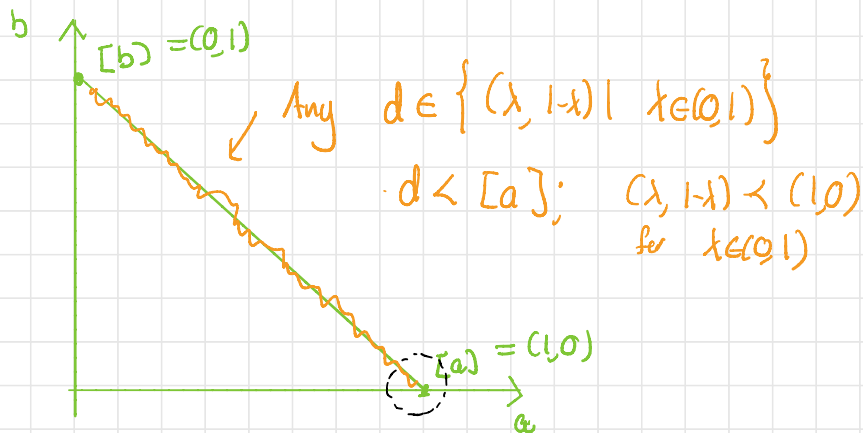
$[a] \sim [b] \sim [c]$

For any  $\alpha \in (0,1)$

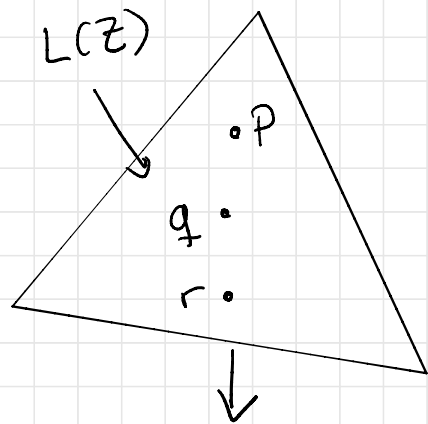
$$\alpha[a] + (1-\alpha)[c] = d$$

$$|\text{supp}(d)| = 2 > \begin{cases} |\text{supp}([a])| = 1 \\ |\text{supp}([b])| = 1 \\ |\text{supp}([c])| = 1 \end{cases}$$

$$d < [b]$$



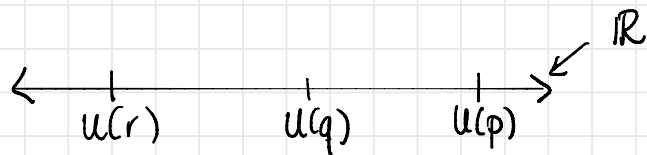
Thus, any ball  $B((1,0))$  will contain lotteries that are indifferent to any point in  $\{(\lambda, 1-\lambda) \mid \lambda \in (0,1)\}$



$$p > q > r$$

The solution is

$$1) \alpha = \frac{u(q) - u(r)}{u(p) - u(r)} > 0$$



"Represent"  $u(q)$  as a convex combination of  $u(r)$  and  $u(p)$

then, to find the weight  $\alpha$

$$\alpha u(p) + (1-\alpha) u(r) = u(q)$$

How do we know  $[M]$  and  $[m]$  exist

$Z$  is finite,  $\{z_1, \dots, z_n\}$

$$z_1 = m, \quad z_n = M$$

$$[z_1] \preceq [z_2] \preceq \dots [z_n]$$

since  $\preceq$  is complete and transitive

$$[z_1] = [m], \quad [z_n] = [M]$$

Any lottery  $p = p(z_1)[z_1] \oplus p(z_2)[z_2] \oplus \dots \oplus p(z_n)[z_n]$

Let  $\alpha(p) = \sum_{z \in Z} p(z) v(z)$  then

$$p \sim \alpha(p) \cdot M \oplus (1 - \alpha(p)) \cdot m$$

$$q \sim \alpha(q) \cdot M \oplus (1 - \alpha(q)) \cdot m$$

By (Mon)

$$p \succeq q \Leftrightarrow$$

$$\alpha(p) M \oplus (1 - \alpha(p)) m \succeq$$

$$\alpha(q) M \oplus (1 - \alpha(q)) m$$

$$\Leftrightarrow$$

$$\alpha(p) \geq \alpha(q)$$

$\Leftrightarrow$  by definition

$$\sum_{z \in Z} p(z) v(z) \geq \sum_{z \in Z} q(z) v(z) //$$

Positive affine transformation:

for any  $u: L(Z) \rightarrow \mathbb{R}$

the transformation  $f: \mathbb{R} \rightarrow \mathbb{R}$

defined by  $f(x) = \alpha x + \beta$ ,

$\alpha > 0$ , is affine and positive

$$f(u) = \alpha u(p) + \beta$$

for  $p \in L(Z)$